Chapter 9
Lungs and Breathing

Abstract The discussion of the human respiratory system begins with a study of lung structure. This is followed by analyzing the microphysics of the lung alveoli and the macrophysics of the lung and the breathing process. Breathing under unusual conditions is also discussed, including breathing with a diseased lung and at high elevations.

Our lungs serve several important functions. They interact with blood by exchanging carbon dioxide for oxygen (Chap. 8) and they maintain the blood pH. The lungs are involved in heat exchange and fluid balance in the body, because relatively dry and usually cooler air is inhaled and air at the body temperature saturated with water vapor is exhaled (Chap. 6). They are also a key element in voice production (Chap. 10). We will highlight the mechanics of breathing [4, 5, 10–12, 14, 22, 24, 26, 27, 34, 41, 42].

We typically breathe in ∼6 L/min of air, which is the *ventilation* or *inhalation* rate. (This is comparable to the ∼5–6 L of blood pumped per min in the pulmonary circulation through the lungs, which is largely by coincidence.) Because air is ∼20% oxygen, we inhale (inspire) ∼1.2 L oxygen/min. The *breathing rate* is typically 12/min for men, 20/min for women, and 60/min for infants. The air we inhale has ∼80% N_2/20% O_2 (or more precisely 78.084% N_2/20.947% O_2/0.934% Ar/0.04% CO_2 for dry air in 2015), and the air we *exhale* (*expire*) has ∼80% N_2/16% O_2/4% CO_2, excluding water vapor and argon. (If the air we exhaled had little or no oxygen, we could not use it for mouth-to-mouth resuscitation.) We breathe in roughly 10 kg air/day, with ∼2 kg O_2/day. The lungs absorb about 0.5 kg O_2/day (400 L). We exhale air with ∼0.5 kg water vapor/day.

Because we inspire and expire air at the same rate (if not where would the difference go?), it is clear the body uses ∼0.2–0.3 L oxygen/min during usual sedentary activity, delivered by the ∼5–6 L of blood pumped per min. We have called this rate of oxygen consumption in the body dV_{O_2}/dt in Chaps. 6 and 8. During aerobic exercise dV_{O_2}/dt increases linearly with cardiac output Q_t (see Fig. 8.26, and (6.18) and (8.28)). The maximum rate of oxygen usage is ∼2.8 L/min for a person of average fitness and ∼4 L/min for a highly fit person. This assumes the lungs bring in air at a rate fast enough to maintain the needed oxygenation of arterial blood. With advancing age, the main factor in decreasing athletic performance is decreasing oxygen intake. At maximal exercise in endurance-trained men, oxygen consumption

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(divided by body mass) decreases by ∼28% from ∼68 mL/kg-min at 28 years of age to ∼49 mL/kg-min at 60 years of age [36].

Gauge pressures, relative to atmosphere, are usually used in discussing breathing. Two roughly equal types of units are commonly used, mmHg and cmH_2O, with 1 mmHg = 1.36 cmH_2O.

9.1 Structure of the Lungs

Air is inhaled through the nose or mouth and then through the pharynx, larynx, and the trachea (windpipe) (Fig. 9.1). The trachea divides into the right and left bronchus (Fig. 9.2), each of which continues to bifurcate into smaller and smaller

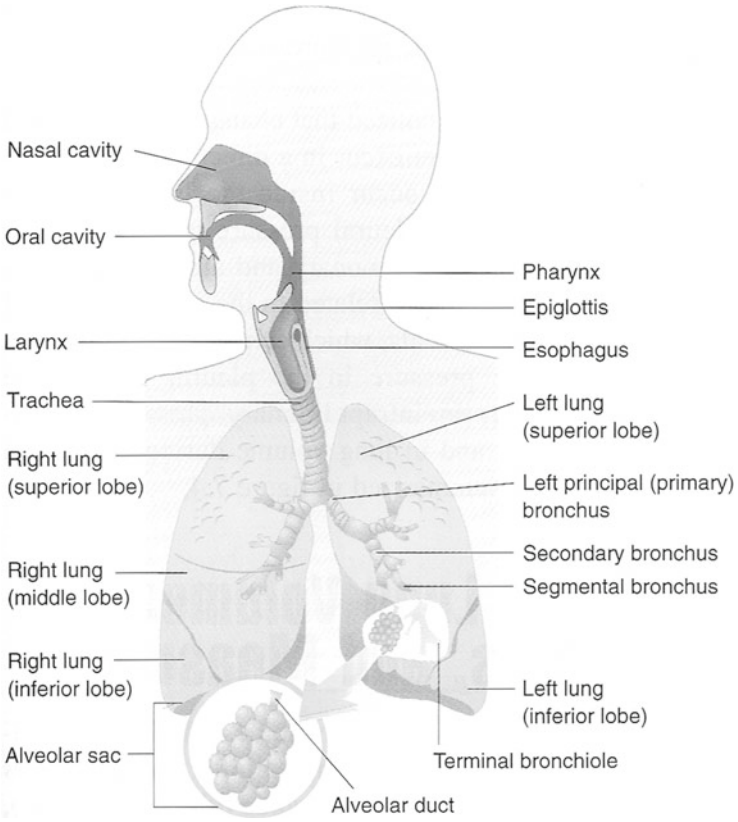


Fig. 9.1 Diagram of parts of the respiratory system. (These components are also important in voice production (Chap. 10). The vocal cords (or vocal folds) used in speaking are in the larynx.) (From [6]. Used with permission)

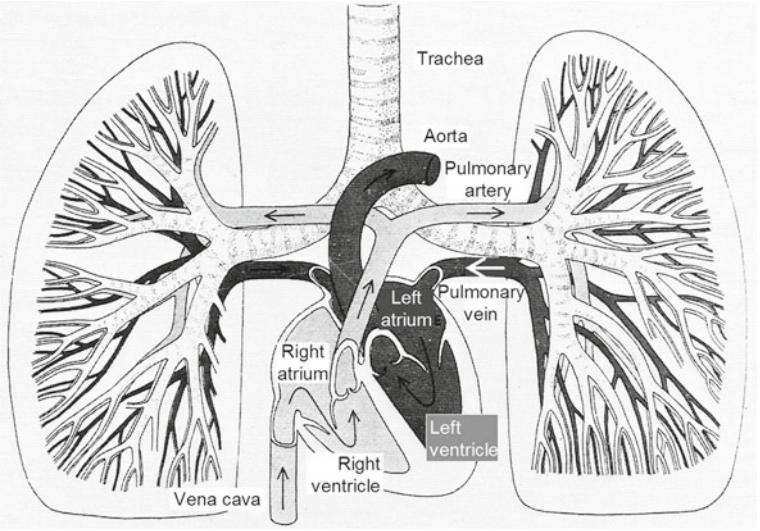


Fig. 9.2 The relationship between the lung and heart is shown. The first few generations of the branching of the air vessels in the lungs, pulmonary arteries, and the pulmonary veins are shown. These three systems can be called the three "trees" of the lung. Note that the pulmonary arteries are close to the bronchi, while the pulmonary veins stand alone (From [9])

bronchi and bronchioles over 23 levels of bifurcation ($2^{24} = 1.6 \times 10^8$) (Table 9.1, Figs. 9.2 and 9.3) until they form alveoli (which is the plural of alveolus) (al-vee-oh'-lie (lus)), which are the actual operating units of the lungs. The average diameter of the airways decreases with generation z, as $d(z) = 2^{-z/3}d(0)$ until generation 16. This relation is the optimal design of a branched system of tubes in hydrodynamics. There are about 3×10^8 alveoli, each ~0.2–0.3 mm in diameter, with walls that are ~0.4 μm thick. They are in contact with blood in the pulmonary capillaries (Fig. 9.4), which themselves form after subdividing in 17 branches (Table 9.2, Figs. 9.2 and 9.5). Oxygen diffuses from the alveoli to the red blood cells, while carbon dioxide diffuses from the blood into the air in the alveoli. The total surface area of the alveoli is ~80 m^2 (ranging from 50–100 m^2). The total external surface area of the lungs is only ~0.1 m^2, so subdividing into alveoli results in a tremendous increase in the surface area in contact with the blood, by a factor of almost 1,000. This is also the factor by which the oxygen intake increases. Without this, we would never even come close to meeting our metabolic needs for oxygen. Our chests expand when we breathe because incoming air filling the alveoli makes each one bigger, just as with ordinary bubbles. Models of O_2 and CO_2 diffusion across the alveoli and pulmonary capillary walls and how this progresses along the capillaries can be found in [13, 14, 28], along with models on how these gases are transported between the alveoli and the rest of the lungs (*ventilation*) and how the capillaries deliver blood to tissue (*perfusion*).

Table 9.1 Approximate quantification of the bronchial system

Pulmonary branch	Generation z	Branch diameter (mm)	Branch length (mm)	Total cross-sectional area (cm^2)	Volume (cm/s)	Air speed (cm^3)
Trachea	0	18.0	120.0	2.5	31	393
Main bronchus	1	12.2	47.6	2.3	11	427
Lobar bronchus	2	8.3	19.0	2.1	4.0	462
	3	5.6	7.6	2.0	1.5	507
Segmental bronchus	4	4.5	12.7	2.5	3.5	392
	5	3.5	10.7	3.1	3.3	325
Bronchi	6	2.8	9.0	4.0	3.5	254
w/cartilage in wall	7	2.3	7.6	5.1	3.8	188
	8	1.86	6.4	7.0	4.4	144
	9	1.54	5.4	9.6	5.2	105
	10	1.30	4.6	13	6.2	73.6
Terminal bronchus	11	1.09	3.9	20	7.6	52.3
	12	0.95	3.3	29	9.8	34.4
Bronchioles	13	0.82	2.7	44	12	23.1
w/muscle in wall	14	0.74	2.3	69	16	14.1
	15	0.66	2.0	113	22	8.92
Terminal bronchiole	16	0.60	1.65	180	30	5.40
Respiratory bronchiole	17	0.54	1.41	300	42	3.33
Respiratory bronchiole	18	0.50	1.17	534	61	1.94
Respiratory bronchiole	19	0.47	0.99	944	93	1.10
Alveolar duct	20	0.45	0.83	1,600	139	0.60
Alveolar duct	21	0.43	0.70	3,200	224	0.32
Alveolar duct	22	0.41	0.59	5,900	350	0.18
Alveolar sac	23	0.41	0.50	12,000	591	0.09
Alveoli, 21 per duct		0.28	0.23		3,200	

Using data from [8, 38]. Also see [39, 40]
The air speed is assumed to be 1 L/s. The data include that for both lungs. The number in each generation is 2^z (for generations $z = 0$–23), and 300×10^6 for the alveoli

The circulatory system is the conduit for the transfer of O_2 and CO_2 between the alveoli and tissues, and so we should track the partial pressure in each system. Within the alveoli the partial pressure of O_2 is $\simeq$105 mmHg, which is smaller than that in the atmosphere (159 mmHg = 21% of 760 mmHg) because of the dead volume in the respiratory system. The partial pressure of O_2 in blood in the pulmonary capillaries increases from 40 to $\simeq$100 mmHg after O_2 is transferred from the alveoli, and this is the partial pressure in the pulmonary veins and systemic arteries. The partial pressure of O_2 in tissue is 40 mmHg, so that after transfer of O_2 from the capillaries to surrounding tissues, the partial pressure in the systemic veins and pulmonary arteries is also $\simeq$40 mmHg—and then it is again increased to 100 mmHg in the lungs.

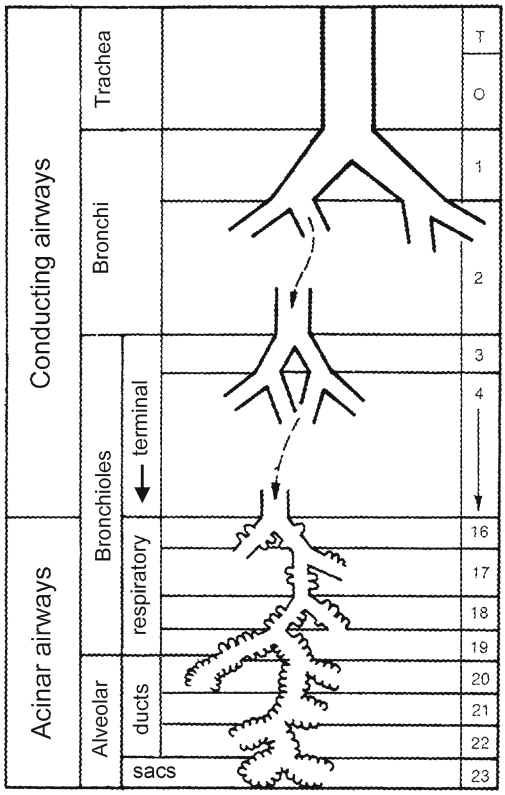


Fig. 9.3 Bifurcations of lung airways, showing generation number z (From [30])

Similarly, within the alveoli the partial pressure of CO_2 is $\simeq$40 mmHg; this is much larger than that in the atmosphere ($\sim$0.3 mmHg), again because of the dead volume. The partial pressure of CO_2 in blood in the pulmonary capillaries decreases from 46 to $\simeq$40 mmHg after CO_2 is transferred to the alveoli, and this is the partial pressure in the pulmonary veins and systemic arteries. The partial pressure of CO_2 in tissue is 46 mmHg, so that after transfer of CO_2 into the capillaries from the tissues, the partial pressure in the systemic veins and pulmonary arteries is also $\simeq$46 mmHg—and then it is again decreased to 40 mmHg in the lungs. With advancing age, the difference in arterial and venous O_2 levels decreases, which indicates less oxygen is being used by the body. At maximal exercise in endurance-trained men, this difference decreases by $\sim$8% from $\sim$16.7 mL(O_2)/100 mL(blood) at 28 years of age to $\sim$15.2 mL/100 mL at 60 years of age [36].

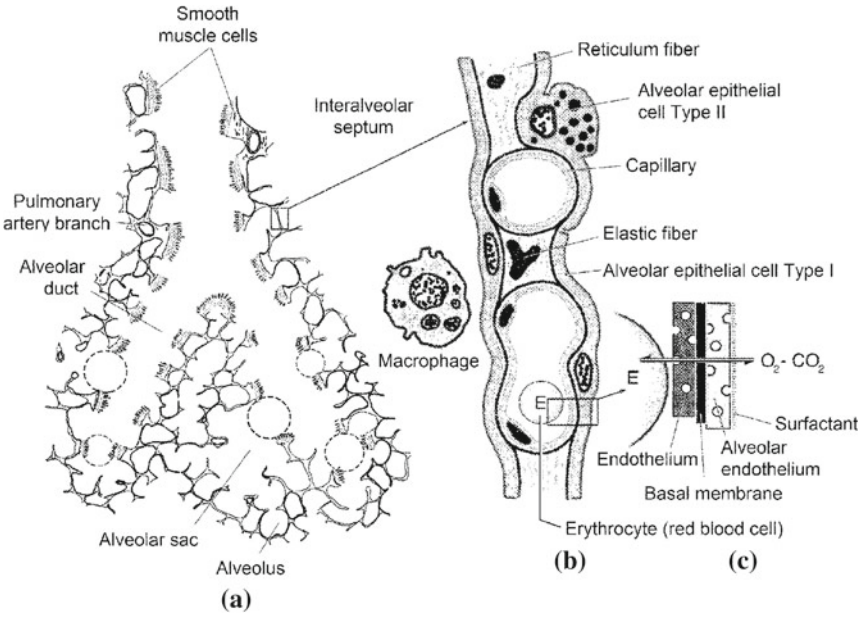


Fig. 9.4 The details of the alveolar bifurcation are shown in (**a**). These alveoli are sacs imbedded in capillary beds. The details of the interaction between the alveoli and capillaries are depicted in (**b**) and (**c**) (From [30])

9.2 The Physics of the Alveoli

The alveoli are similar to interconnected bubbles. Inside them the pressure is P_{in} and outside the pressure is P_{out}, with $\Delta P = P_{\text{in}} - P_{\text{out}}$, and they have a radius R. The Law of Laplace for a sphere (7.9) is

$$\Delta P = \frac{2T}{R}, \tag{9.1}$$

where T is the tension in the sphere walls. The main source of this tension in the alveoli is not within the walls but on the surfaces. This contribution is called the surface tension γ, which has the same units as T—of force/length or energy/area. In typical bubbles, such as soap bubbles, both surfaces contribute the same surface tension and so T is replaced by 2γ. Therefore we find

$$\Delta P = P_{\text{in}} - P_{\text{out}} = \frac{4\gamma}{R}. \tag{9.2}$$

For the water/air interface $\gamma \simeq 7.2 \times 10^{-4}$ N/m (Table 7.2). In alveoli, however, only the surface tension of the inner surface is really important because it is a fluid/air

Table 9.2 Branching structure of the pulmonary arterial network

Pulmonary branching order	Number of branches of each order	Vessel length (mm)	Vessel diameter (mm)
1	1	90.5	30.0
2	3	32.0	14.83
3	8	10.9	8.06
4	20	20.7	5.82
5	66	17.9	3.65
6	203	10.5	2.09
7	675	6.6	1.33
8	2,290	4.69	0.85
9	5,861	3.16	0.525
10	17,560	2.10	0.351
11	52,550	1.38	0.224
12	157,400	0.91	0.138
13	471,300	0.65	0.086
14	1,411,000	0.44	0.054
15	4,226,000	0.29	0.034
16	12,660,000	0.20	0.021
17	300,000,000	0.13	0.013

Using data from [8, 33]

interface and has larger surface tension than the fluid/fluid interface of the outer surface, and so

$$\Delta P_{\text{alveoli}} = P_{\text{in}} - P_{\text{out}} = \frac{2\gamma}{R}. \tag{9.3}$$

There is an apparent instability that seemingly leads to an unreasonable situation in interconnected bubbles or alveoli. Consider two bubbles that are initially not interconnected, as in Fig. 9.6, because there is a plug between them. Bubble #1 has an internal pressure P_1 and radius R_1, and Bubble #2 has an internal pressure P_2 and radius R_2. (Because the difference between the pressure inside and outside the bubble is what is significant, the external pressure is equal to zero.) In equilibrium, the internal pressure $P_{\text{internal}} = 4\gamma/R$ for each bubble. (Whether this factor is 4 or 2 is not significant here.) Say Bubble #2 is the smaller bubble. Because $R_2 < R_1$, in equilibrium $P_2 > P_1$; the smaller bubble has the higher internal pressure. If the plug is opened, air will flow from higher pressure to lower pressure, and therefore from the smaller bubble to the larger bubble. The loss of air in Bubble #2 makes it smaller. With this smaller radius, the equilibrium internal pressure increases. Because this pressure is still higher than in Bubble #1, air continues to flow from the smaller bubble to the larger bubble, until it collapses.

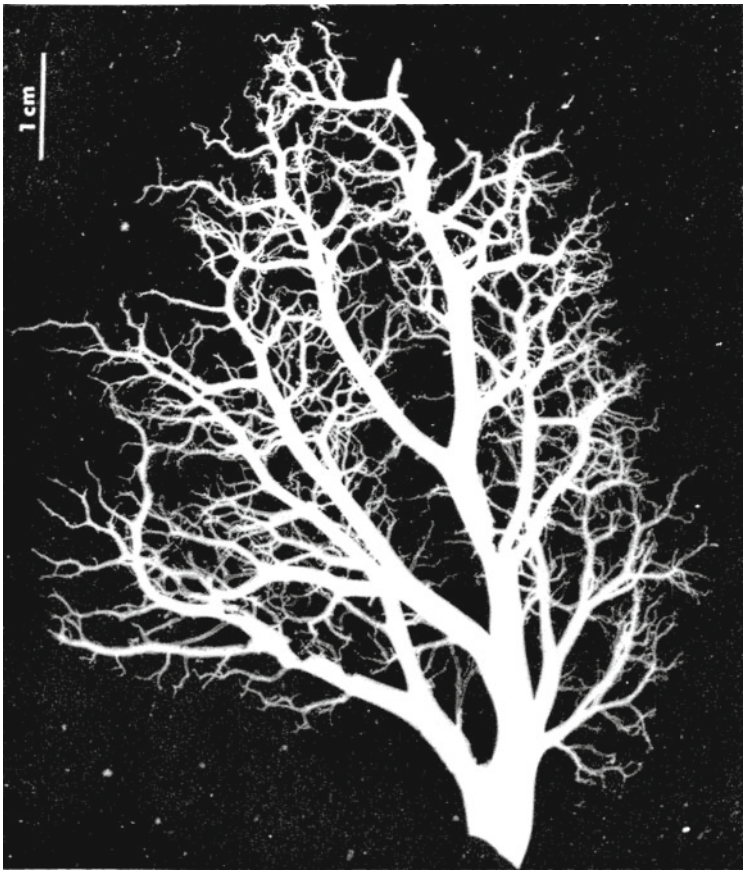


Fig. 9.5 A silicone elastomer cast of the venous tree of the lung of a cat. The venous pressure was $-7\,\mathrm{cmH_2O}$ ($= -5\,\mathrm{mmHg}$), the airway pressure was $10\,\mathrm{cmH_2O}$ ($= 7\,\mathrm{mmHg}$), and the pleural pressure was $0\,\mathrm{cmH_2O}$ (From [9])

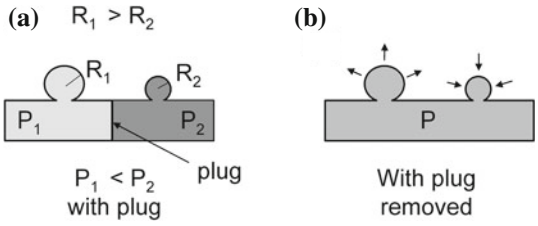


Fig. 9.6 Instability of bubbles, according to the Law of Laplace. This assumes that the surface tension does not change with bubble (or alveolus) radius. The external pressure is 0

This implies that the largest of the hundreds of millions of alveoli would get ever larger at the expense of all of the smaller ones and the system of alveoli we have described for the lungs could not be stable. What is wrong with this scenario? There is no error in our reasoning; however, we have made one assumption that is not accurate

for alveoli. We implicitly assumed that the surface tension is not a function of radius R. There is a surfactant on the surfaces of the alveoli of healthy people, containing dipalmitoyl phosphatidycholine or DPPC, that causes $\gamma(R)$ to be independent of R from 0 to a critical size (R_0). With $\Delta P = 2\gamma(R)/R$, some bubbles become bigger and others smaller as in Fig. 9.6. However, eventually $\gamma(R)$ increases with larger R faster than R does itself, so ΔP begins to increase with even larger R. Such a system of interconnected alveoli is stable.

We can see how such a dependence of $\gamma(R)$ can occur with the following model. The surface of an alveolus can be covered either with a lipoprotein or by water; the surface tension of the lipoprotein is much lower ($\gamma_{\text{lung}} = 1 \times 10^{-3}$ N/m) than that of water ($\gamma_{\text{water}} = 7.2 \times 10^{-2}$ N/m (= 72 dynes/cm)). Assuming the alveolus is spherical, for one particular radius R_0 there is exactly one monolayer of lipoprotein on the whole surface and at that radius the surface tension is γ_{lung} over the $4\pi R_0^2$ surface area. If this alveolus becomes smaller, so $R < R_0$, it has several monolayers of lipoprotein on its surface and its surface tension is still γ_{lung}, and so

$$\gamma(R) = \gamma_{\text{lung}} \quad \text{for } R \leq R_0. \tag{9.4}$$

If this same alveolus instead becomes larger, so $R > R_0$, it has a monolayer of lipoprotein over only a portion of its surface (of surface area $4\pi R_0^2$ because the layer cannot become smaller than a monolayer) and water over the rest of the surface (of area $4\pi R^2 - 4\pi R_0^2$). So the average surface tension is

$$\gamma(R) = \frac{4\pi R_0^2 \gamma_{\text{lung}} + (4\pi R^2 - 4\pi R_0^2)\gamma_{\text{water}}}{4\pi R^2} \quad \text{for } R > R_0 \tag{9.5}$$

or

$$\gamma(R) = \gamma_{\text{water}} - \frac{R_0^2}{R^2}(\gamma_{\text{water}} - \gamma_{\text{lung}}) \quad \text{for } R > R_0. \tag{9.6}$$

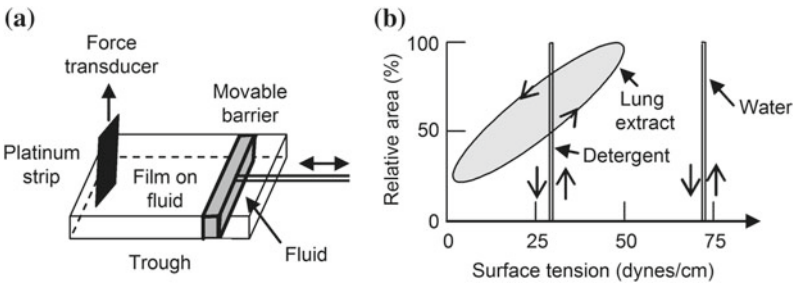


Fig. 9.7 Surface tension on alveoli walls (lung extract) in (**b**), as measured by the surface balance in (**a**) which measures surface tension versus area. Similar measurements for detergent and water are also shown (Based on [12, 42])

This approaches the much larger γ_{water} for $R \gg R_0$ at a rate that is faster than R, so the alveoli will be stable.

Because this lipoprotein is only on one of the surfaces, the stability condition is $\Delta P = 2\gamma/R$. For $R > R_0$, there is a stable equilibrium when $\mathrm{d}(\Delta P)/\mathrm{d}R = \mathrm{d}[2\gamma_{\text{water}}/R - 2(R_0^2/R^3)(\gamma_{\text{water}} - \gamma_{\text{lung}})]/\mathrm{d}R = 0$ or

$$\frac{\mathrm{d}(\Delta P)}{\mathrm{d}R} = -\frac{2\gamma_{\text{water}}}{R^2} + 6\frac{R_0^2}{R^4}(\gamma_{\text{water}} - \gamma_{\text{lung}}) = 0 \tag{9.7}$$

or

$$R_{\text{eq}} = \sqrt{3\frac{\gamma_{\text{water}} - \gamma_{\text{lung}}}{\gamma_{\text{water}}}}\, R_0. \tag{9.8}$$

Because $\gamma_{\text{water}} \gg \gamma_{\text{lung}}$, the equilibrium radius $R_{\text{eq}} \simeq \sqrt{3}R_0$.

Figure 9.7 shows that this surface tension of the surfactant in the lung decreases from 5×10^{-2} N/m (50 dynes/cm) to zero as the area of the film gets smaller. Alveoli are typically stable at approximately 1/4 of their maximum size.

One function of the surfactant is to provide alveolus stability. Another function is to lower the amount of force needed to be supplied by the diaphragm to inflate the alveoli. With $\gamma_{\text{water}} = 7.2 \times 10^{-2}$ N/m and $R = 0.05$ mm of the alveoli when they are collapsed (and need to be inflated), (9.3) gives $\Delta P_{\text{alveoli}} = 2.9 \times 10^3\,\text{N/m}^2 = 22$ mmHg. The area of an adult diaphragm muscle is about 500 cm^2, so the force it needs to exert to expand the alveoli for breathing is ~150 N—which corresponds to a weight with mass 15 kg. With the lower surface tension of the lung surfactant, this force is over an order of magnitude smaller and breathing is easier, especially for infants. This explains why people with insufficient surfactant—with hyaline membrane disease—have difficulty breathing.

9.3 Physics of Breathing

Each lung is surrounded by a sac membrane within the thoracic cavity. We can picture the pleural sac as a balloon, as in Fig. 9.8, filled with intrapleural fluid. The inside wall of this sac, the visceral pleura (membrane), attaches to the outer lung wall. The outside wall of this sac, the parietal pleura (membrane), attaches to the thoracic wall. It is the springiness of the lung that pulls the two pleural membranes apart, and this causes a slight decrease of pressure of the pleural sac relative to atmospheric pressure of −4 mmHg to −6 mmHg. This pressure difference is what keeps the lungs expanded, and keeps them from collapsing. The mechanical "driving force" in controlling lung volume is the *transpulmonary pressure*, which is the difference in pressure in the alveoli in the lungs and that around the lung in the pleural sac, which is called the *intrapleural (or pleural) pressure*. (The alveolar and pleural pressures are gauge pressures, referenced to atmospheric pressure.)

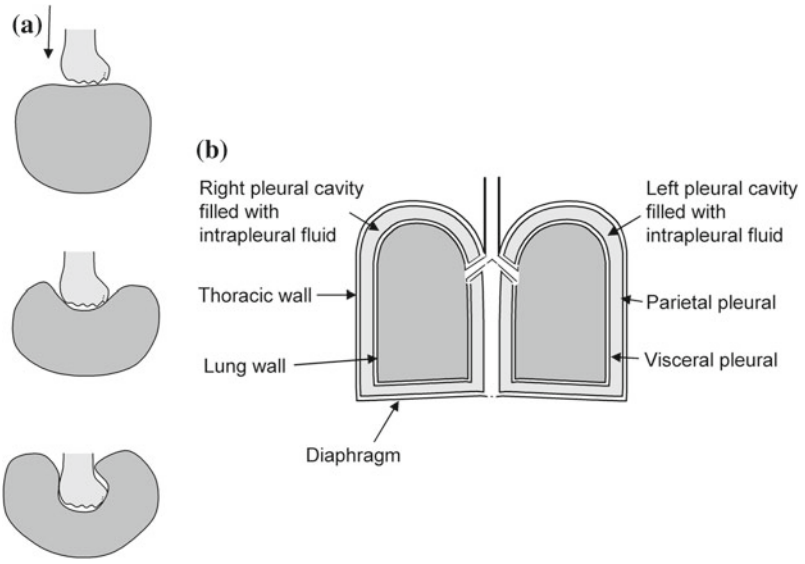


Fig. 9.8 **a** Pushing a fist into a balloon is analogous to the lungs in the pleural cavity. **b** Schematic of the lungs in the pleural cavities (Based on [34])

The lungs are expanded and contracted by the motion of structures surrounding them by way of inspiratory and expiratory muscles. This occurs in two ways (Fig. 9.9), of which only the first one is used during quiet breathing. (1) The diaphragm moves downward to lengthen the chest cavity (by pulling the bottom of the lungs downward) during inspiration. During quiet breathing, the lungs contract by the natural elastic recoil of the lungs and chest wall, with the diaphragm relaxed, while in heavier breathing this contraction is accelerated by the contraction of the abdominal muscles that push the abdominal contents and then the diaphragm upward to shorten the chest cavity. (2) The ribs are elevated by the neck muscles to increase the anteroposterior (front-to-back) diameter of the chest cavity and are depressed (lowered) by the abdominal recti to decrease it. This causes chest cavity expansion and contraction, respectively, because the ribs slant outward and have larger transverse cross-sectional areas in the lower sections; this can increase the anterior–posterior chest thickness by about 20% during inspiration.

How does this help bring air into the lungs? Before inspiration, there is atmospheric pressure in the lungs. The attractive force of the visceral pleura for the parietal pleura and the outward force of the outer lung wall due to the lower-than-atmospheric pressure in the pleural sac (∼–4 mmHg) cause each lung to expand. In equilibrium their sum is balanced by the tendency of the lungs to contract due to their springiness. This preinspiration force balance is shown in Fig. 9.10. They are no longer in balance during inspiration.

The steps in inspiration (inhaling) are shown in Fig. 9.11. The inspiratory muscles (diaphragm and external intercostals) increase the dimensions of the rib cage (the

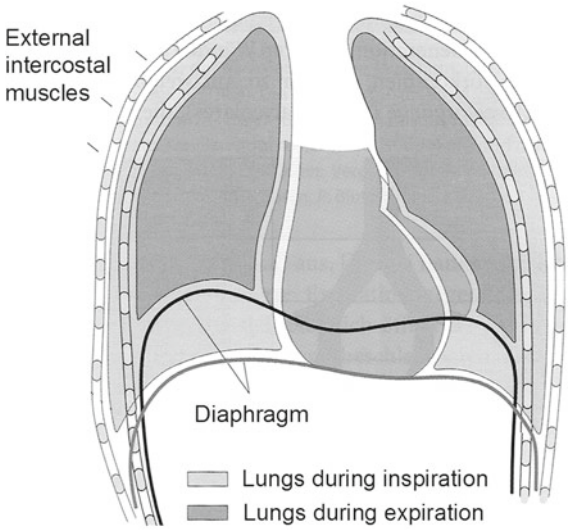


Fig. 9.9 Expansion and contraction of the thoracic cage during expiration and inspiration, showing the ribs, lungs and heart, the external intercostal muscles (that contract during inspiration to elevate the rib cage and widen it laterally so the cage increases in all three dimensions), and the diaphragm (that contracts to increase the vertical dimension of the cage during inspiration) (From [31])

thoracic cavity). This causes the visceral and parietal pleurae to separate. The lung volume then increases because (1) the attraction of the visceral and parietal pleurae increases as they are separated further and (2) this separation causes $P_{lung} - P_{pleura}$ to decrease even more, from ~–4 to ~–6 mmHg (i.e., from ~756 to 754 mmHg absolute pressure). Because both of these forces in the direction of lung expansion increase, they now overcome the springiness of the lungs that favors lung contraction—and the lung expands. The pressure in the lungs and alveoli decreases from

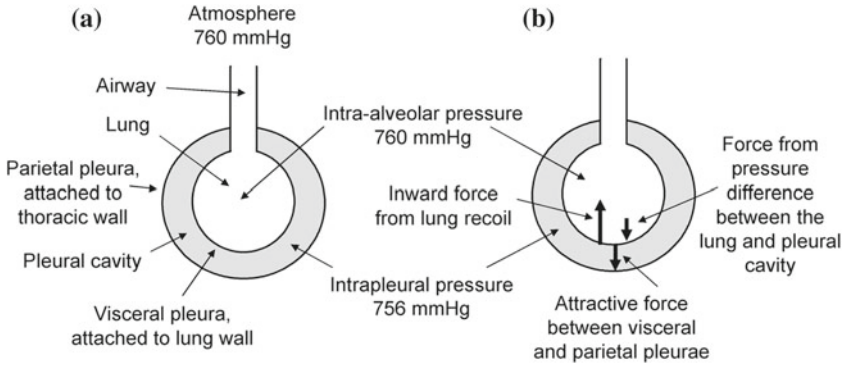


Fig. 9.10 Force balance of the visceral pleura/outer lung wall during preinspiration. Note that the forces are really normal to the wall everywhere, not just at the bottom as depicted (Based on [34])

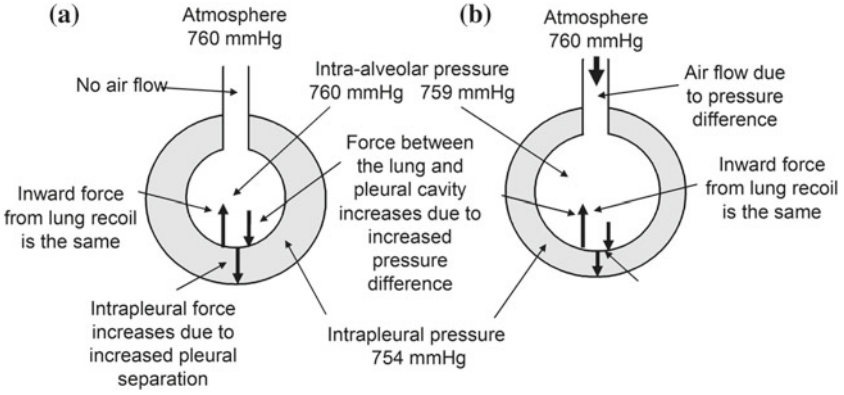


Fig. 9.11 **a** Force imbalance at the onset of inspiratory muscle contraction (and expansion of the thoracic wall/parietal pleura) leads to a (**b**) subatmospheric pressure in the lungs and flow of air into the lungs during inspiration. (Compare this to the preinspiration force balance in Fig. 9.10) (Based on [34])

$\sim$0 to $\sim$–1 mmHg (i.e., from $\sim$760 to 759 mmHg absolute pressure), and then air flows from the mouth and nose into the lungs. During normal breathing exhaling is automatic, requiring no contraction by muscles. Muscle contraction is necessary during heavy exercise to inhale more fresh air and to actively exhale stale air.

9.4 Volume of the Lungs

The volume of the lungs depends on the transpulmonary pressure, as is seen in Fig. 9.12 for an excised lung. The inflation and deflation curves are not the same; as in the Chap. 4 discussion of viscoelasticity, this asymmetry is called hysteresis.

The volume of the lungs during different stages of normal and deep breathing is a good diagnostic of lung functionality. It is easily measured using a *spirometer* (Fig. 9.13). Figure 9.13 shows one such measurement during different types of breath-

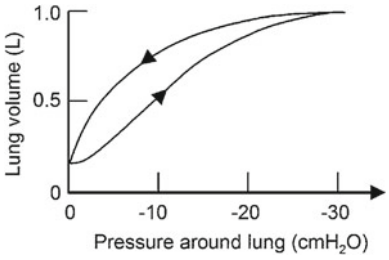


Fig. 9.12 The measurement of the pressure–volume curve of an excised lung, which shows hysteresis in inflation and deflation (Based on [42])

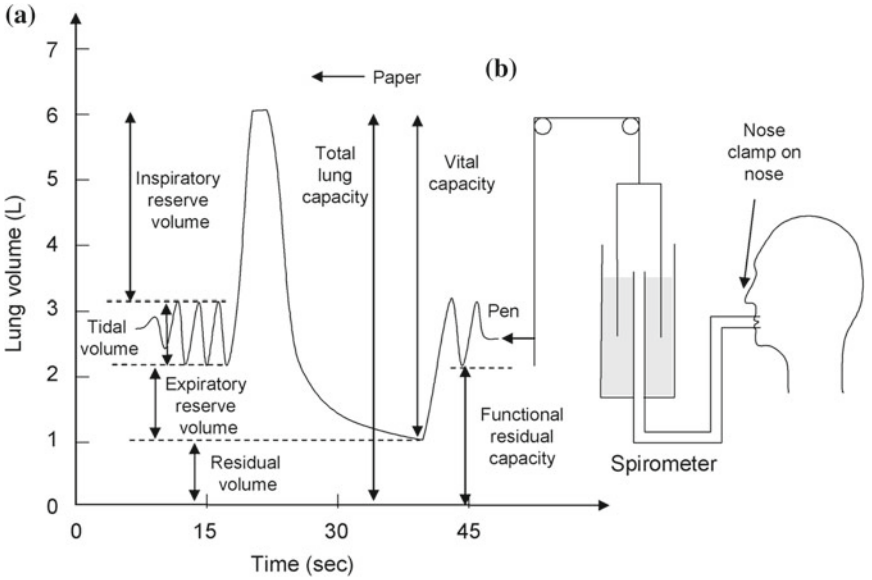


Fig. 9.13 **a** Lung volume changes during breathing cycles, **b** along with a schematic of a spirometer (Based on [3, 42])

ing. In this example, during normal breathing the lung volume is seen to oscillate between 3.2 L after normal inspiration and 2.2 L after normal expiration. The difference is the tidal volume (TV) (~1 L), which is the usual lung volume used during breathing when at rest. The volume after normal expiration is the functional residual capacity (FRC). After a deep inspiration the lung volume is the total lung capacity (TLC), ~6 L. This exceeds the volume after normal inspiration by the inspiratory reserve volume (IRV, which is also one of the author's nicknames). After a deep expiration, the remaining volume is the residual volume (RV), ~1 L. The difference in lung volumes after deep inspiration and deep expiration is the vital capacity (VC) ~5 L, which also equals the total lung capacity minus the residual volume. After deep expiration the lung volume is smaller than that after normal expiration by the expiratory reserve volume (ERV), which also equals the functional residual capacity minus the residual volume.

During pregnancy the inhalation rate increases by ~30–50%, due to the increased tidal volume and respiratory frequency [19]. Oxygen consumption increases by ~20–40% to meet the needs of the fetus and placenta, and additional needs of the mother's organs.

The vital capacity is an important measure of how well the lungs are functioning. (More importantly, you need a robust vital capacity to be able to inflate balloons.) The functional reserve capacity is the volume of stale air that normally mixes with new air (the tidal volume). There is also dead space. Some is anatomic (0.15 L), due to the trachea and bronchii, and some is physiological alveoli dead space, where the alveoli have no access to blood. With advancing age, the total lung capacity does

not change, but the vital capacity decreases. Alveolar dead space increases, which affects the oxygen level in the arteries, but not the elimination of carbon dioxide [32, 37].

We know that we inhale air that is ~80% N_2/20% O_2/0% CO_2, again ignoring water vapor and Ar, it mixes with stale air, and after diffusion across the alveoli walls we exhale air that is ~80% N_2/16% O_2/4% CO_2. *What is the composition of this stale air?* Let us call its composition 80% $N_2/x\%$ $O_2/y\%$ CO_2. After inspiration the lungs have a tidal volume of 80% N_2/20% O_2/0% CO_2 air that has mixed with a functional reserve capacity of 80% $N_2/x\%$ $O_2/y\%$ CO_2 air. If the tidal volume is a fraction α of this volume, the functional reserve capacity volume fraction is $1-\alpha$. After inhalation the composition of air in the lungs is 80% $N_2/[20\alpha + (1-\alpha)x]\%$ $O_2/(1-\alpha)y\%$ CO_2. Say a fraction β of all of the inhaled air is absorbed by the lungs. This means that a fraction β of the $20\alpha\%$ O_2 in the lungs is absorbed, which is $20\alpha\beta\%$ O_2. There is an equal $20\alpha\beta\%$ increase in CO_2 that diffuses into the lungs. So, after the oxygen and carbon dioxide transfer, the air in the lungs has a composition of 80% $N_2/[20\alpha - 20\alpha\beta + (1-\alpha)x]\%$ $O_2/[20\alpha\beta + (1-\alpha)y]\%$ CO_2, which is exhaled. Therefore, the oxygen and carbon dioxide fractions are, respectively,

$$16\% = [20\alpha - 20\alpha\beta + (1-\alpha)x]\% \tag{9.9}$$

$$4\% = [20\alpha\beta + (1-\alpha)y]\%. \tag{9.10}$$

The tidal volume fraction is $\alpha = 1/3$ and the fraction of inhaled oxygen that is absorbed by the lungs is $\beta = 1/4$. This gives $x = 16.5$ and $y = 3.5$, so the stale air in the lungs has a composition 80% N_2/16.5% O_2/3.5% CO_2. (This air is clearly oxygenated enough to be useful during mouth-to-mouth resuscitation.)

9.5 Breathing Under Usual and Unusual Conditions

9.5.1 Flow of Air During Breathing

During inspiration, air flows because the pressure is lower in the lungs and alveoli by a positive amount ΔP than in the atmosphere. The amount of air that flows is determined by the resistance and compliance of the respiratory system.

The compliance of the lung is ~0.2 L/cmH_2O and it decreases for a normal person with higher expanding pressures, as is seen by the decreasing slope in Fig. 9.12. In trying to evaluate lung performance, the *specific compliance* of the lung is perhaps more meaningful, for which the lung compliance is normalized by a characteristic of the person's size, such as a characteristic lung volume (FRC, VC, TLC, etc.), the lung dry weight, or the body weight. The elasticity of the chest, as well as that of the lung, contributes to the lung compliance.

Airway resistance is dominant within the generations of the intermediate sized bronchii, as seen in Fig. 9.14a and Problem 9.21. Poiseuille's Law (7.24) can be used to calculate the airway resistance in the lungs (Problems 9.21–9.23) and other passages (Problems 9.16 and 9.18). The total airway resistance is typically $\sim$2 $cmH_2O/(L/s)$ during normal breathing and it decreases with increasing lung size (Fig. 9.14b). Respiratory resistance increases during pregnancy [19]. (It is measured as in Fig. 9.17 below.)

This airway resistance is about 80% of the total *pulmonary resistance*. The other 20% is due to viscous forces in chest and lung wall movement, and this is called *tissue resistance*.

The elastic properties of the thoracic cage (with the chest well) are important in breathing, as are those of the lungs. Figure 9.15 shows the volume of the lungs, chest walls alone, and the lungs in the chest wall as a function of pressure. This is measured after inspiration or expiration with a spirometer and subsequent relaxing of respiratory muscles. At every volume the pressure (the relaxation pressure) of the lung/chest wall combination is the sum of those for the lungs and chest walls separately.

With advancing age, the lung compliance seems to increase in healthy people, but the compliances of the chest wall and the total respiratory system decrease (and the latter increases the work of breathing, as we will see below). There are no apparent changes in airway resistance [32, 37].

What are the maximum expiratory and inspiratory pressures? For healthy young men, the average gauge pressure developed during expiration is $\sim$100–150 cmH_2O ($\sim$0.097–0.145 atm) and during inspiration it is ~ -95 to -105 cmH_2O (which is also called an inspiration pressure of $\sim$95–105 cmH_2O), and for healthy young women it is $\sim$60–95 cmH_2O during expiration and $\sim$–65 to -75 cmH_2O during inspiration [15, 44]. There is a very wide range in these average maximum pressures (even

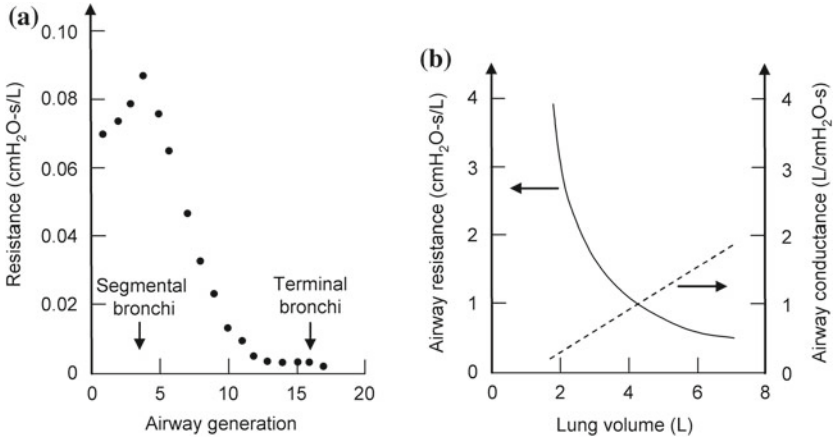


Fig. 9.14 **a** Airway resistance for each bronchus generation, and **b** total airway resistance and conductance versus lung volume (Based on [42], from **a** [25], **b** [2])

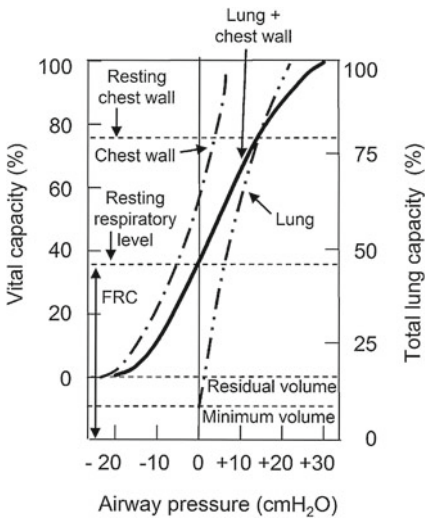


Fig. 9.15 Pressure–volume curves of the lungs, chest wall (no lungs), and chest wall with lungs, with relaxed respiratory muscles. The lung data are the same as those in Fig. 9.12, except no hysteresis is shown and the airway pressure is of the opposite sign to the pressure around the lung. FRC is the functional residual capacity. The measurement is made with a spirometer-like apparatus, similar to that in Fig. 9.13b (Based on [42])

much outside the range reported here), depending on the age group and the specific measurement method, and in addition there is a large spread among individuals of the same group even when the same method is used. In any case, these extreme expiratory and inspiratory pressures are respectively very useful for blowing up balloons and sipping fluids up very long straws. (What is the longest drinking straw you can use? See Problem 9.27.)

Is the pressure difference between the alveoli and atmosphere large enough to drive the right amount of air into our lungs each breath? (Under normal conditions, it had better be.) During each breath, this pressure difference starts at zero, increases to a maximum $\Delta P_{\max}$, and then decreases to zero again at the end of the breath. Let us determine the average pressure difference in this sequence. We will model inhalation as a half cycle of a sine wave with: $\Delta P(t) = \Delta P_{\max} \sin(2\pi f t) = \Delta P_{\max} \sin(\pi t / T_{\text{half period}})$, which lasts a half-cycle time $T_{\text{half period}} = 1/(2f)$. (The parameter f is the same as the breathing or respiratory rate only if the inhalation and exhalation times are the same.) For $f = 0.25$ Hz, $T_{\text{half period}} = 2$ s. The average pressure difference during this inhalation is

$$\Delta P_{\text{av}} = \frac{1}{T_{\text{half period}}} \int_0^{T_{\text{half period}}} \Delta P_{\max} \sin(\pi t / T_{\text{half period}}) \mathrm{d}t \tag{9.11}$$

$$\begin{aligned} \Delta P_{\text{av}} &= \frac{\Delta P_{\max}}{T_{\text{half period}}} \frac{T_{\text{half period}}}{\pi} \left(\cos(\pi T_{\text{half period}} / T_{\text{half period}}) - \cos(0)\right) \\ &= \frac{2}{\pi} \Delta P_{\max}, \end{aligned} \tag{9.12}$$

so for $\Delta P_{\max} = 1.1$ mmHg, we see that $\Delta P_{\text{av}} = 0.7$ mmHg.

Let us say that the inflow of air per breath is V_{in}. Then the average flow rate is $Q_{\text{av}} = V_{\text{in}}/T_{\text{half period}}$. For $V_{\text{in}} = 0.5$ L, this is 0.25 L/s. If the flow rate is proportional to the pressure drop, Q and ΔP have the same dependence on time and so $Q_{\max} = (\pi/2)Q_{\text{av}}$, which is $(\pi/2)0.25\,\text{L/s} = 0.4\,\text{L/s}$, and $Q(t) = Q_{\max}\sin(2\pi f t) = Q_{\max}\sin(\pi t/T_{\text{half period}})$. Moreover, $Q = \Delta P/R_{\text{flow}}$, where R_{flow} is the total resistance to flow in the nasal passages, trachea, and so on. The resistance to flow is $R_{\text{flow}} = \Delta P/Q$, and so using average values we see that $R_{\text{flow}} = 0.7\,\text{mmHg}/(0.25\,L/s) = 3.7 \times 10^5$ Pa-s/m^3.

Is the flow laminar or turbulent? If the trachea has a radius of 9 mm, the air flows at a maximum speed $u_{\max} = Q_{\max}/A = (400\,\text{cm}^3/\text{s})/(\pi(0.9\,\text{cm})^2) = 160\,\text{cm/s}$. The Reynolds number (7.11) is $Re = \rho u d/\eta$. Using the mass density of air at body temperature $\rho = 1.16 \times 10^{-3}\,\text{g/cm}^3$ and the air viscosity 2×10^{-5} Pa-s $= 2 \times 10^{-5}(\text{N/m}^2)\text{s}$, we find that $Re = (1.16 \times 10^{-3}\,\text{g/cm}^3)(160\,\text{cm/s})(1.8\,\text{cm})/(2 \times 10^{-4}(\text{dyne/cm}^2)\text{s}) = 1{,}600$, so the flow would generally be expected to be laminar in the trachea, as well as in the inner nasal passages and the pharynx. However, some turbulence is expected because the walls of these passages are not smooth.

Air flow through the nose is also interesting. Because our noses project outside the skull with nostrils at the bottom, the air we inspire hits the inside of the nose and changes its path by 90°. During this turbulent flow to the inner nose, air hits mucus membranes that line the inside of the nose and the low-humidity inspired air becomes saturated with water, which is necessary so the lungs do not dry out. The flow through the nose also cleans the air through nose hairs and helps warm the air before it reaches the lungs. This added resistance does, however, make the lungs work harder (see below). Turbulence during expiration helps the membranes recapture much of the air water vapor, so the body does not lose much water during resting and also during periods of great activity, when the air flow rates are much larger [7, 16].

9.5.2 Mechanical Model of Breathing and Model Parameters

We have just examined only the resistance to flow during breathing. Figure 9.16b shows a more complete mechanical model of the lungs and breathing. It is a compliance vessel described by $V(t) = V_{\text{d}} + C_{\text{flow}}P(t)$, attributed to the springiness of the lungs, in series with an inertial element. They are in parallel with a resistive element, attributed to the airway resistance we just examined. These model elements are driven by a pressure determined by the inspiratory muscles.

9.5.3 Inspiration/Expiration Cycle

Figure 9.17 shows the lung pressure, rate of flow of air into the lungs, and lung volume versus time during a cycle of inspiration and expiration. If the effect of airway

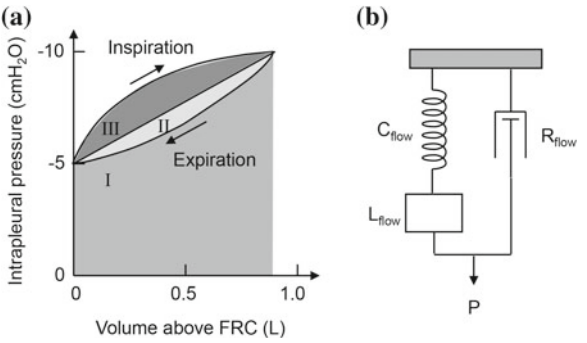


Fig. 9.16 **a** Work done during inspiration (areas I + II + III) and work recovered during expiration (area I). With no viscous, resistive forces, the work in inflating the lung would be areas I + II, and this is associated with the lung compliance. The extra work done overcoming respiratory flow resistance is area III. **b** Mechanical model of breathing has the lumped compliance (elastance) C_{flow}, resistance R_{flow}, and inertance L_{flow}, and P represents the inspiratory muscles. The inertance can be neglected except for large flows (Based on [12, 41])

resistance were neglected, the alveolar pressure would be zero and the intrapleural pressure would follow the broken curve, which is determined by the elastic recoil of the lung.

It is also instructive to plot these variables as functions of each other. Lung volume is plotted versus lung pressure for all times during a breathing cycle in the dashed cycle trajectories in Fig. 9.18. Time is an implicit variable along the trajectories. Such plots are useful because the model of the lungs includes a compliance vessel in which volume and pressure are interrelated. The differences in various modes of breathing are easily seen in such plots.

9.5.4 Breathing with a Diseased Lung

The static and dynamic pressure–volume curves in Fig. 9.18 indicate how different diseases affect lung compliance, lung volume, and airway resistance. *Obstructive* disorders are due to airway obstructions that increase airway resistance, and limit exhalation at rapid breathing rates. They include chronic bronchitis (excessive mucus production in the bronchial tree), emphysema (enlargement of air spaces after the terminal bronchiole, with the destruction of respiratory system walls, such as those of the alveoli), and asthma (widespread narrowing of airways, sometimes spontaneously). Such obstructions (Fig. 9.19) can be due to excessive secretions (caused by chronic bronchitis), thickening of airway walls (edema or muscle hypertrophy), and outside abnormalities (edema, enlarged lymph nodes, or destruction of lung alveoli tissue as in emphysema). (Edema is an excessive accumulation of fluid in tissue spaces or a body cavity.) These obstructions and the loss of small airways due to the destruction of lung tissue all increase airway resistance. The breakdown of

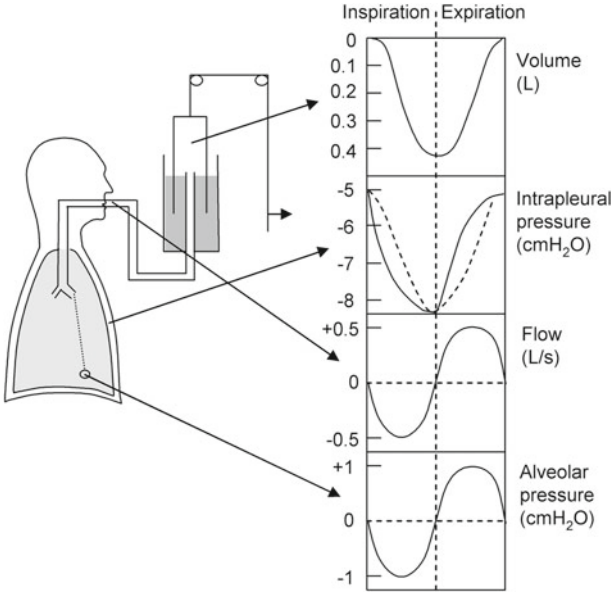


Fig. 9.17 Lung volume, intrapleural pressure, flow, and alveolar pressure versus time during a breathing cycle. If the effect of airway resistance were neglected, the alveolar pressure would be zero and the intrapleural pressure would follow the broken curve (Based on [42])

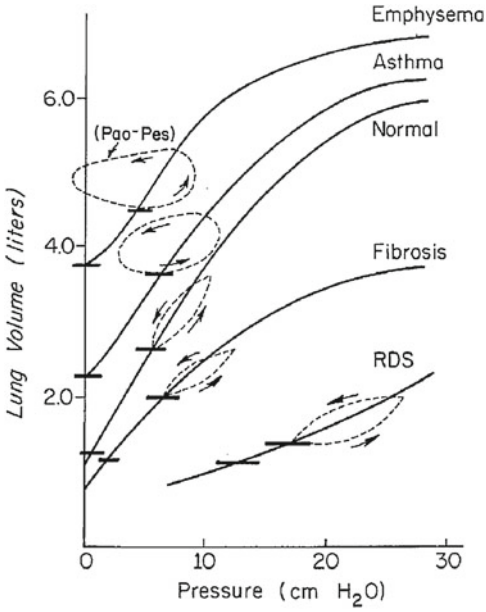


Fig. 9.18 Typical lung volume versus pressure for patients with various respiratory conditions. The monotonically increasing curves are for static conditions, with the lower horizontal bar being the residual volume and the upper horizontal bar the functional residual capacity. Representative dynamic breathing loops (enclosed *dashed curves*) for tidal volume breathing are shown for each condition (Reprinted from [12], with permission of Elsevier)

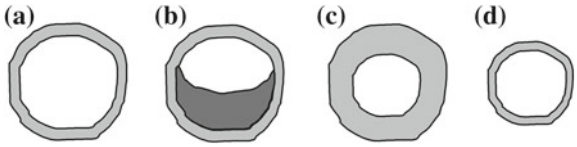


Fig. 9.19 **a** Normal airway and obstructed airways due **b** blocking, **c** airway wall thickening, and **d** outside abnormality. Also see Fig. 9.22 (Based on [41])

elastic alveoli walls also reduces the springiness, and therefore also the compliance. *Restrictive* disorders are those in which the expansion of the lung is restricted. The decreased compliance (slope) in *pulmonary fibrosis* and idiopathic respiratory distress syndrome (RDS) is clear in Fig. 9.18. They are characterized by a lower vital capacity, but airway resistance (per lung volume) is not increased.

Poor breathing due to a diseased lung is manifest in different ways. As seen in Fig. 9.20, the inspired volume is very low if the compliance is less than normal and/or the airway resistance is greater than normal. A spirometer with a low resistance can also be used to test for lung malfunctions in a manner that is slightly different from that used in Fig. 9.13. After a very deep inhalation, a person forces air out as fast as possible. The total volume exhaled is the forced vital capacity (FVC), which can be a bit less than the vital capacity measured with slower expiration. Also of note is the volume exhaled in the first 1 s, which is the forced expiratory volume (FEV) (or FEV_1, which denotes specifically that this volume was expired in 1 s), and the ratio FEV/FVC. As seen in Fig. 9.21, for a normal lung FVC $=$ 5.0 L and FEV/FVC $=$ 80%. The example of an *obstructive* pattern has a lower FV, 3.1 L, and smaller FEV/FVC, 42%. The example of a *restrictive* pattern also has a lower FV, 3.1 L, but a high FEV/FVC, 90%. The flow rates for the obstructive pattern are also abnormally low (Problem 9.31).

Figure 9.18 shows cycles that typify the breathing cycles of a normal person and of people with lung disorders. The divisions between the alveoli break down in people

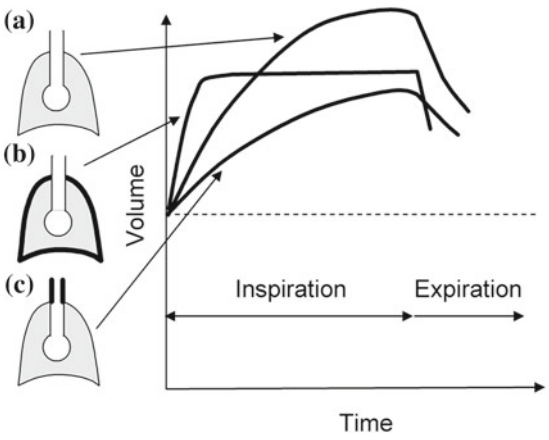


Fig. 9.20 Lung volume during inspiration for **a** normal conditions, **b** decreased compliance, and **c** increased airway resistance (Based on [42])

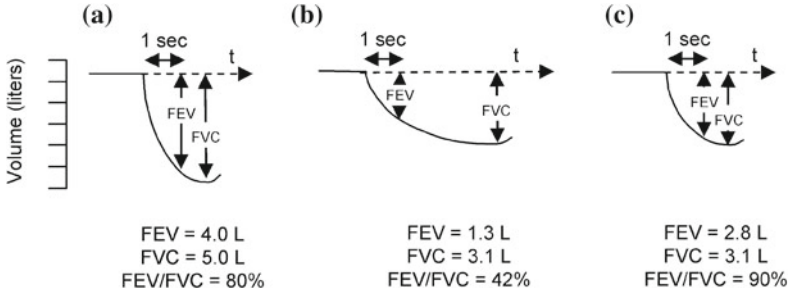


Fig. 9.21 Forced expiration for **a** normal, **b** obstructive, and **c** restrictive patterns (Based on [41])

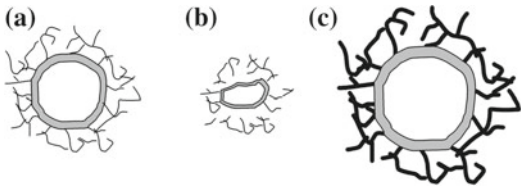


Fig. 9.22 Airway dimensions for different patients (Based on [41]). **a** Normal, **b** Emphysema, **c** Fibrosis

with *emphysema*. Consequently, the lungs become less springy and more compliant, and the airway resistance contribution dominates breathing. In pulmonary fibrosis, the compliance is reduced by an increase in fibrous tissue. This condition increases in pulmonary emphysema and in normal aging, due to a change in elastic tissue in the lungs. The volume/pressure locus is also shown for idiopathic respiratory distress syndrome (RDS). It is seen to move to higher pressure due to a lack of alveoli surfactant, as occurs in some premature babies. In such infants the minimal surface tension is only 2×10^{-4} N/m, compared to $\ll 0.5 \times 10^{-4}$ N/m for normal lungs. With less surfactant, the alveolus surface tension decreases and, at the same pressure difference the alveolus is smaller. Figure 9.22 shows that the airways tend to collapse in emphysema because of the loss of radial traction with exterior structures, while in interstitial fibrosis the airways can become large in diameter, due to excessive radial traction, making the airways large in volume relative to the lungs.

9.5.5 Breathing at Higher Elevations and Under Other Unusual Conditions

The partial pressure of oxygen is 161 mmHg (21.2 kPa) at sea level (300 K), which corresponds to $\simeq$20.9% O_2 content. Healthy people begin suffering adverse health effects as the O_2 fraction dips increasingly below 19.5% at normal sea level atmosphere [23, 29]. As this fraction decreases to 10%, the body responds by increasing breathing rates (tachypnea) and heart beat rates (tachycardia) to increase oxygen consumption, as it suffers increasingly impaired attention, thinking, and coordination; these responses are triggered at increasingly low levels of exertion. Breathing

air with oxygen fractions in range 6–10% results in nausea, vomiting, lethargic movements, and perhaps unconsciousness. Breathing air containing less than 6% oxygen produces convulsions, then cessation of breathing (apnea), followed by cessation of ventricular contractions and ejection of blood by the heart (cardiac standstill). This critical threshold is called *hypoxia* (which is the lack of oxygen reaching living tissues) and occurs below a critical partial pressure of $\simeq$57 mmHg (7.5 kPa) for a normal, relaxed breathing rate. (Another way of saying all of this is: Our bodies have been designed so that our rate of breathing air, rate of transferring oxygen to the blood, the capacity of the blood to hold oxygen, the rates of blood flow to tissues, and so on leads to a metabolism that functions well when the partial pressure of oxygen is reasonably above this critical value.)

These effects can occur at sea level atmospheric pressure when the air supply is not replenished so the oxygen fraction decreases (as in sealed rooms and cars without sufficient makeup fresh air and in bubbles in capsized boats (see Problem 9.42, which also addresses the effect of increased CO_2 fractions)) and when the fraction of oxygen remains essentially normal but the partial pressure of O_2 is lower than normal because of the lower atmosphere pressure at high elevations. (The body can adapt only a bit to long-term decreases in oxygen available at higher elevations.) *At what elevation above sea level does hypoxia occur?*

The pressure of a fluid column of constant fluid density ρ and constant gravitational constant g is given by (2.48), $P = \rho g h$, where h is the height of the column. In Chap. 8 we considered a column of blood; now let us consider a column of air at a height z and above, where $z = 0$ at sea level. The change in pressure for a change in height is

$$\mathrm{d}P = -\rho g \mathrm{d}z. \tag{9.13}$$

Because we will be considering heights above sea level that are much smaller than the radius of the Earth, we have ignored the dependence of the gravitational acceleration g on z. (This is considered in Problem 9.43.)

The ideal gas law (7.2) is $P = nRT$, where n is the density, R is the gas constant, and T is the temperature, or

$$P = \frac{RT}{m}\,\rho, \tag{9.14}$$

where $\rho = mn$ is the mass density, with m the molecular mass. We will assume that the atmospheric temperature (300 K) does not vary with height (see Problem 9.48). Dividing (9.13) and (9.14) gives

$$\frac{\mathrm{d}P}{P} = -\frac{mg}{RT}\,\mathrm{d}z. \tag{9.15}$$

Integrating from sea level, $z = 0$ with pressure $P_{\text{sea level}}$, to a height h, with pressure $P(h)$, gives

$$P(h) = P_{\text{sea level}} \exp\left(-\frac{mg}{RT}\,h\right). \tag{9.16}$$

Using the partial pressure of oxygen at sea level and $m = 32$ g/mol for oxygen, this shows that hypoxia occurs at a height of 8.25 km (= 27,100 ft = 5.1 miles). (The height of Mt. Everest is 29,029 ft.)

9.6 Work Needed to Breathe

During inspiration the thorax and abdomen do positive work to expand the lungs [1, 12, 18, 42]. The body does no work during normal expiration, but during forced breathing work is also done to contract the lungs during expiration. Because $V = V_d + C_{flow}P$, for the breathing cycle with volume changing between the functional residual capacity volume, V_{FRC}, and that plus the tidal volume, $V_{FRC} + V_t$, we can write $V(t) = V_{FRC} + C_{flow}P(t)$. We see that $\Delta V(t) = V(t) - V_{FRC} = C_{flow}P(t)$ or $P(t) = \Delta V(t)/C_{flow}$. The work done during inhaling a tidal volume V_t is

$$W = \int_0^{V_t} P\mathrm{d}(\Delta V) = \int_0^{V_t} \frac{\Delta V}{C_{flow}}\mathrm{d}(\Delta V) = \frac{V_t^2}{2C_{flow}}. \tag{9.17}$$

In Fig. 9.16a, the work done overcoming these elastic (compliance) effects is the area defined by regions I + II, and this is what we have derived here (and will use below). The work is really larger, the area represented by regions I + II + III, because of viscous (resistive) effects. (These viscous/resistive effects also lead to the hysteresis here and in Fig. 9.12, just as in Chap. 4.)

For a breathing rate of f, the rate of doing work for inspiration is

$$\frac{\mathrm{d}W}{\mathrm{d}t} = f\frac{V_t^2}{2C_{flow}}. \tag{9.18}$$

With a breathing rate of 20/min, tidal volume of 500 cm^3, and lung compliance of 0.1 L/cmH_2O = 0.1 $cm^3/(dyne/cm^2)$, this is

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{(20/\mathrm{min})(500\ \mathrm{cm}^3)^2}{2 \times 0.1\ \mathrm{cm}^3/(\mathrm{dyne}/\mathrm{cm}^2)} = 3.6 \times 10^3\ \mathrm{J/day} = 0.86\ \mathrm{kcal/day}. \tag{9.19}$$

The respiratory muscle efficiency is $\epsilon = 5$–10%, and so the metabolic need is

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{\mathrm{d}W/\mathrm{d}t}{\epsilon} = \frac{0.86\ \mathrm{kcal/day}}{0.05} = 17\ \mathrm{kcal/day} \tag{9.20}$$

assuming 5% efficiency; this calculation is very sensitive to the values chosen for C_{flow} and ϵ. This result is about 1% of the BMR; however this value is really about 2% of the BMR, so maybe ϵ is closer to 3% and/or other effects need to be considered, such as dissipation due to resistance of the flow and the viscous nature (of the overall

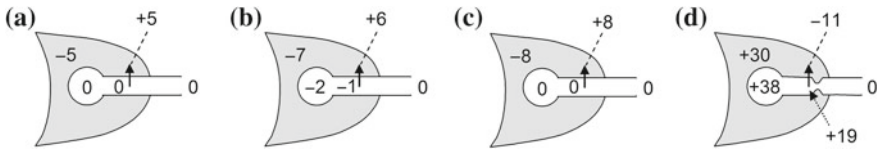


Fig. 9.23 Intrapleural, alveoli, airway, and atmospheric pressures in cmH_2O during inspiration and forced expiration (Based on [42]). **a** Preinspiration, **b** During inspiration, **c** End-inspiration, **d** Forced expiration

viscoelasticity) of the lungs and chest wall. Airflow resistance in the nose seems to be responsible for about half of the work needed to breath.

During normal breathing there is no mechanical work done in expiration. During heavy workouts and strenuous exercise the metabolic needs increase because there is (1) also work done during expiration, (2) a faster breathing rate, and (3) a larger tidal volume. The work done to breathe can then use 25% of the total body energy consumption. Such a large increase is suggested by Fig. 8.26. Rapid shallow breathing (as recommended in Lamaze training for childbirth) requires more energy than does normal breathing, to overcome the resistive nature of flow in the air passages. Similarly, slow, deep breathing requires more energy to overcome the elastic nature of the lung and chest. Problem 9.12 compares other relative advantages and disadvantages of these last two modes of breathing.

Another interesting thing can occur during forced expiration. In normal breathing the pressure in the airways always exceeds the intrapleural pressure during inspiration and expiration (as is seen in Fig. 9.23 during normal inspiration). However, during forced expiration the intrapleural and alveoli pressure both increase by the same amount ($38\,cmH_2O$) and so they are both positive relative to the atmosphere (+30 and $+38\,cmH_2O$ in the figure). There is then a large pressure drop from the alveoli to the lips and nostrils. At some point along this route the pressure in the airway will be lower than $30\,cmH_2O$ and this will compress the airways (shown where the pressure is $+19\,cmH_2O$ in this figure) and increase the airway resistance.

9.7 Summary

The macroscopic basis of lung function and breathing can be understood by analyzing the volume, pressure, and air flow during breathing, by using models of fluid flow and lung expansion. The physical nature of the individual operating units in the lungs, i.e., the alveoli, can also be understood this way. These models can also be used to understand the physical consequences of a diseased lung. The energy needed to operate the lungs can also be modeled.

Problems

Lungs

9.1 Calculate the effective lung volumes and breathing rates for a man (70 kg), woman (50 kg), and an infant (5 kg) using the allometric relation parameters in Table 1.13. How do the breathing rates compare with those given above?

9.2 If there are 3×10^8 alveoli in a lung with a functional residual capacity (FRC) of 2.5 L, calculate the average volume and radius of an alveolus.

9.3 Use Table 9.1 to show that the air travels a total distance of 273 mm from the trachea to the alveoli.

9.4 What is the total volume of the lungs described in Table 9.1? Where is most of the volume?

9.5 Is continuity of flow obeyed by the data for the lungs in Table 9.1? Check this using the data for bronchial generations 0, 1, 2, 3, 4, 5, 10, 16, 20, and 23.

9.6 Calculate the Reynolds number for the bronchial generations listed in Problem 9.5. Is the flow laminar or turbulent in the respiratory system?

9.7 Calculate the pressure drop across pulmonary arterial orders 1, 4, 10, 13, 16, and 17, assuming a total blood flow of 5 L/min.

9.8 The CO_2 level in the atmosphere was ~280 ppm (parts per million) in preindustrial times and is ~380 ppm now (2015). Express these levels in terms of mmHg. Would this change be expected to affect the exchange of CO_2 in the lungs in any significant manner?

Alveoli and Surface Tension

9.9 Derive (9.8) from (9.7).

9.10 Estimate the force the adult diaphragm would need to exert if there were no lung surfactant.

Breathing

9.11 Use Fig. 9.13b to explain how a spirometer works. How much should the water in the spirometer rise and fall during breathing cycles? (Assume reasonable dimensions for the instrument.)

9.12 During breathing, the pulmonary ventilation, V_p (in L/min) (the rate at which air enters the trachea), equals the respiratory rate, R (in units of per min), times the tidal volume, V_t (in L). Because of the anatomical dead space volume V_d, only $V_t - V_d$ enters the alveoli (and is thus of use). Therefore, a more meaningful ventilation rate is the alveolar ventilation $V_a = R(V_t - V_d)$:
(a) Find V_p and V_a for the conditions in Table 9.3, assuming $V_d = 0.15$ L.
(b) Compare the pulmonary ventilation for the four breathing patterns in this table for the person at rest. (Patterns (ii)–(iv) require more metabolic power than does (i), because of increased work due to resistance to flow and resistance in the tissues for (ii) and (iii), and increased work due to compliance (elastic) forces of the lung and chest in (iv).) Which of the four are clearly inadvisable because of poor alveolar ventilation?
(c) During exercise, both the respiratory rate and tidal volume increase. Based on the results in part (a) for (i)–(iv) and for (v)–(vi), do you gain more by breathing faster or deeper for a given pulmonary ventilation?

9.13 What are the maximum and average air flows for each breathing cycle in Problem 9.12, assuming the inhalation and exhalation periods are the same?

9.14 (a) When you take in a deep breath of say 1 L, how much does your mass (in kg) and weight (in N and lb) increase?
(b) Does your average density increase, decrease, or stay the same? If there is a change, estimate it.

9.15 (a) What does Fig. 8.26 say about the amount of oxygen that can be consumed per amount of cardiac output?
(b) What does it say about how much oxygen is needed to do work? Is this consistent with what is presented in the text?
(c) How is work output defined in this figure?

9.16 Estimate the resistance of the trachea using Poiseuille's Law, assuming it has a radius of 9 mm and a length of 110 mm. How does this compare to the total resistance?

9.17 Estimate the resistance of the vocal tract using Poiseuille's Law, assuming it can be modeled as three tubes in series with respective lengths 6, 3, and 6 cm and cross-sectional areas 5, 1, and 5 cm^2. (Also sketch this model.)

Table 9.3 Examples of breathing cycles

Activity		R (per min)	V_t (L)
(i)	At rest, quiet breathing	12	0.5
(ii)	At rest, with rapid, shallow breathing	24	0.25
(iii)	At rest, with very rapid, very shallow breathing	40	0.15
(iv)	At rest, with slow, deep breathing	6	1.0
(v)	Exercising, with rapid, shallow breathing	24	0.5
(vi)	Exercising, with slow, deep breathing	12	1.0

9.18 Estimate the resistance of the nasal passage using Poiseuille's Law, assuming it has a radius of 4 mm and a length of 3 cm. How does this compare to the total resistance and is it a limiting factor in the resistance to flow?

9.19 If you model the breathing airway as a series of sequential passages, the nasal or mouth passage, the pharynx, larynx, and then trachea, each with a resistance to air flow, what is the total resistance to air flow in terms of these individual resistances?

9.20 In both inspiration and expiration, a pressure difference of 0.4 cmH_2O causes a flow of 0.15 L/s in the nose. Determine the flow resistance in it.

9.21 Consider the lung bifurcation generations 1–19 in Table 9.1:
(a) In which generation is the flow resistance largest? What is its value?
(b) Do your results agree with those in Fig. 9.14a?
(c) In which generation is the pressure drop greatest, and generally in what range of bifurcations is most of the pressure drop?

9.22 The total airway resistance is the sum of those in each lung generation. Do the resistances in Fig. 9.14a add to give you a total resistance consistent with that in Fig. 9.14b?

9.23 Calculate the resistance for lung generation 4 using Poiseuille's Law and compare it to the values given in the chapter.

9.24 In blowing up a balloon you exhale 80% of the vital capacity (of 4 L) with each breath. How many breaths are needed to blow up a balloon from a very small (ignorable) volume to a sphere with a diameter of 10 inches?

9.25 (a) When inflating a balloon, the volume initially increases slowly with increasing pressure until a maximum pressure is reached (when the balloon is only partially inflated) and as you continue to blow the volume increases quickly, and the internal pressure decreases. Explain why some people can easily inflate balloons and others cannot.
(b) In one study, the volume of the balloon was increased slowly until the internal gauge pressure reached 2.0×10^4 dyn/cm^2 and the volume then was increased more rapidly, as the internal pressure decreased to 1.5×10^4 dyn/cm^2 [17, 20]. Would you expect to be able to blow up this balloon (and why do you think so)?

9.26 Not everyone has the same maximum expiratory pressure. Suppose the maximum expiratory pressure is 100 ± 40 cmH_2O for a group of men and 60 ± 40 cmH_2O for a similar group of women. Estimate the fraction of people that are capable of blowing up a given type of balloon, for which the inflation pressure is 80 cmH_2O. Assume there are equal numbers of men and women.

9.27 (a) What is the longest vertical (drinking) straw a man with a maximum inspiratory pressure of 150 cmH_2O can use to drink water (using one breath)? (Ignore limitations due to inspiratory volume and resistive flow considerations.)
(b) What is the largest (inside) diameter the straw could have, given the limitation that the person can maintain this pressure only for 40% of the vital capacity of 4.5 L?

9.28 (a) Use Fig. 9.15 to determine the compliance of the lungs ($C_{\text{flow,lung}}$) and chest walls ($C_{\text{flow,chest wall}}$) at 0, 20, 40, 60, and 80% of vital capacity. (b) Determine the compliance of the combined lung/chest wall system ($C_{\text{flow,lung/chest wall}}$) at these volumes, and compare these values with those from part (a) by using $1/C_{\text{flow,lung/chest wall}} = 1/C_{\text{flow,lung}} + 1/C_{\text{flow,lung/chest wall}}$.

9.29 (a) Use Fig. 9.18 to determine the compliance of the lungs for each condition (within the lowest 5 cmH_2O pressure range shown for each).
(b) How does the compliance vary for each over the pressure range shown?

9.30 Compare the specific lung compliances of a 65 kg man and 20 g mouse, with respective compliances of 0.2 L/cm-H_2O and 0.0001 L/cm-H_2O.

9.31 Show that the forced expiratory flow (FEF) rates for the normal, obstructive, and restrictive flows in Fig. 9.21 are 3.5, 1.4, and 3.7 L/s, respectively. Do this by determining the slopes of the three curves in this figure. (Use a straight-line fit between points that have decreased by 25 and 75% on the way to the FVC.)

9.32 Determine the air flow resistance from the flow rate and alveoli pressure in Fig. 9.17.

9.33 Consider only the compliance in the work of breathing and assume that the compliance C_{flow} for normal lungs is 0.1 cm^5/dyne:
(a) In fibrosis of the lungs the compliance of the lungs decreases. For a given tidal volume, how does the rate of work of breathing change if the compliance decreases by x%?
(b) Compare the rate of work done in breathing (J/day) and the associated rate of metabolism (kcal/day) (if the muscles associated with breathing are 5% efficient) for cases (i) and (iv) in Problem 9.12 for normal lungs.

9.34 (advanced problem) Write down the equation of motion for the mechanical model in Fig. 9.16b and solve it for inspiration.

9.35 (advanced problem) Show that the solution in Problem 9.34 qualitatively agrees with the trends seen in Fig. 9.20: with decreased compliance the time constant decreases and the volume breathed during a cycle decreases, while with increased airway resistance the time constant increases and the volume breathed during a cycle decreases.

9.36 Estimate the rate of energy consumed by the lungs during exercise with a breathing rate of 40/min and tidal volume of 1,000 cm^3.

9.37 Use a blood circulation rate of 5 L/min and the known change in the partial pressures of O_2 and CO_2 in the systemic capillaries to find the number of liters of O_2 consumed and CO_2 exhaled each day. How do your results change if you instead use the change in the partial pressures of O_2 and CO_2 in the pulmonary capillaries? Explain why.

9.38 If your chest wall and parietal pleura of a lung are punctured, the intrapleural pressure will increase to atmospheric pressure and that lung will collapse. Explain why. Also draw a diagram explaining this.

9.39 (a) When you engage in activities that require a high metabolic rate, like running, you sometimes breathe through your mouth to increase your rate of air intake. This may occur, for example, when your nostrils flare inward when you try to inspire too quickly through your nose, which impedes air flow. Explain why you can become dehydrated breathing in and out of your mouth and not your nose.
(b) When you run would it be helpful to inspire through your mouth (to increase the rate of air intake) and expire through your nose?

9.40 If all the air you inspire (5 L/min) has zero relative humidity and becomes fully saturated with water (47 mmHg at body temperature) during inspiration, how much water (in L) would you lose each day if none of this water vapor were recaptured during expiration? Could this cause dehydration?

9.41 Let us continue Problem 9.40, by calculating the rate of energy loss by the body due to the loss of water heat of vaporization during inspiration, with no subsequent condensation during expiration. How significant is this?

Breathing at Higher Elevations and Under Other Unusual Conditions

9.42 A boat capsizes and a sailor gets trapped in an air bubble that rises to the top of the overturned boat as he awaits rescue [21, 35].
(a) What volume of fresh air at atmospheric pressure would someone like the sailor inspire in two days? Assume a breathing rate of 6 L air/min. In this problem, give all volumes in terms of m^3 and the length of the cube (in ft) with that volume.
(b) Let us assume the answer to (a) is the volume of this bubble. What fraction of the air would be O_2 after two days if it started at a fraction of 20.9%, the sailor continues to breathe at the usual volumetric rate, and consumes a net of 0.3 L O_2/min (which takes into account the O_2 in expiration)? What is the condition of the sailor after two days?
(c) Recalculate the volume of this bubble after two days, now remembering that one CO_2 molecule is produced for each O_2 molecule consumed. Ignore diffusion of air through the water and the solubility of CO_2 in water.
(d) Show that for every additional 10 m below the water surface, local pressure increases by 1 atm (the value at sea level), and then re-determine the volumes of the bubble in (a) and (c) if the bubble formed when the boat was at sea level and then the boat moved to 30 m below sea level. How could these volumes be significant if the boat were small?
(e) When the CO_2 level in air increases from the usual 400 ppm to about 5% a person becomes confused and panicked, starts hyperventilating, and eventually loses consciousness. What is this percentage in the air bubble after two days? (In any case, if rescued in time the sailor would still need to spend time in a decompression chamber to avoid the "bends" (decompression sickness) if he were indeed 30 m or

more below sea level, because of the increased amount of nitrogen dissolved in the blood at this higher pressure. If this procedure were not performed, nitrogen bubbles would form in his body when he returns to sea level, which could result in his death.)

9.43 (a) Show that the gravitational acceleration constant g varies with height z above sea level as $g(z) = g\,(R_{\text{Earth}}/(R_{\text{Earth}} + z))^2$, where the radius of the Earth is $R_{\text{Earth}} = 6,378\,\text{km}$.
(b) Show that this variation does not affect the analysis of oxygen deprivation at high elevations, described in the text.

9.44 What is the atmospheric pressure in the "mile-high" city of Denver? What is the partial pressure of oxygen there?

9.45 Commercial jets typically cruise at an altitude of ∼10,700 m (∼35,000 ft). What are the total pressure and partial pressure of oxygen at that height? Why are jets pressurized? Why are oxygen masks made available just in case the cabin is depressurized?

9.46 Why do some athletes train at high elevations?

9.47 Apply (9.16) to the variation of the partial pressure of nitrogen, using $m = 28\,\text{g/mol}$. Let us say here that the ratio of oxygen to nitrogen is 20.9%/78.1% = 0.268 at sea level. What is this ratio at the critical height for hypoxia?

9.48 The temperature of the troposphere (the atmosphere up to roughly 11 km) decreases with height, by a bit less than 1 K per 100 m of elevation. In the standard atmosphere $T(z) = T_{\text{sea level}} + \alpha z$, with $T_{\text{sea level}} = 288.19\,\text{K}$ and $\alpha = -0.00649\,\text{K/km}$. (For a dry atmosphere, $\alpha = -0.0098\,\text{K/km}$, than without this consideration.)
(a) Use this temperature variation in (9.15) to show that

$$P(h) = P_{\text{sea level}}\,(T_{\text{sea level}}/(T_{\text{sea level}} + \alpha h))^{gm/R\alpha}\,. \tag{9.21}$$

(b) Show that hypoxia occurs at a lower elevation, 7.21 km, than without this consideration.

9.49 Table 9.4 compares the partial pressure of oxygen (in mmHg) in the air and in the body at sea level and at an elevation of 2,500 m:
(a) Justify the values given for total pressure and O_2 partial pressure at 2,500 m.
(b) Justify the O_2 partial pressure in the alveoli at 2,500 m by using the pressure at sea level.
(c) At sea level your blood flows at a rate of 5 L/min. How fast would it have to flow at an elevation of 2,500 m to provide the same flow of oxygen to the tissues? (Assume no change in the red blood cell and hemoglobin concentrations in the blood. These increase as part of adapting to higher elevations.)
(d) How much faster would you have to breathe at this elevation (in liters of air per min) to maintain the same rate of oxygen delivery? How could this translate into changes in the breathing rate and tidal volume?

Table 9.4 Total and partial pressures at different elevations

Altitude	Sea level	At 2,500 m
Atmospheric pressure (total)	760	560
Atmospheric pressure (O_2)	159	117
In alveoli (O_2)	105	77
In arterial blood (O_2)	100	72
In venous blood (O_2)	40	40

Using data from [43]

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